

Miles Jennings

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EDUCATION

University of California, Irvine

Irvine, CA

B.S Computer Engineering — Deans Honor List — GPA: 3.5

Sept. 2022 – June 2026

Relevant Coursework: Digital Systems, Advanced C Programming, OOP, Data Structures & Algorithms, System Software, and Computer Networks

Organizations: HyperXite, FUSION, FASAE

EXPERIENCE

Test Engineer Intern

Buena Park, California

Leach International Corp.

March 2026 – Present

- Designed and deployed an automated **Python**-based capacitor validation and conditioning system integrating an LCR6002, high-voltage power supply, and safety analyzer, eliminating outsourced testing for 490 components and reducing validation costs by **60%**
- Developed a reusable **LabVIEW** state machine interfacing with a **DAQ's digital I/O** to control a solid-state relay, replacing one-off manual test scripts with a standardized framework for future automated production test systems
- Diagnosed and repaired **8** malfunctioning test systems within **3** months by analyzing undocumented system architectures and troubleshooting **hardware test fixtures**, reducing operator downtime and production bottlenecks for various product lines
- Authored detailed electrical wiring diagrams in **Altium** to re-engineer and restructure key component testing machinery, simplifying **signal routing** and reducing future system troubleshooting time

AudioVisor

Irvine, California

Research for EECS Department @UCI

June 2025 – March 2026

- Led end-to-end development of an audio-reactive wearable device for the hearing impaired, designing the embedded system architecture and leading firmware development.
- Developed DUT firmware in C on the nRF52840 to interface with a digital microphone over PDM, enabling real-time audio sampling and sound direction/amplitude detection

Control Systems Engineer

Irvine, California

HyperXite – SpaceX Hyperloop

Aug. 2025 – June 2026

- Built ESP32-based bidirectional communication system using ESP-NOW for real-time control and telemetry between pod and station and utilizing FreeRTOS to schedule critical tasks
- Evaluated the team's existing embedded architecture against an **STM32H753ZI** solution, selecting the STM32 for lower power consumption and native **CAN/ADC** support, eliminating external interface hardware and improving embedded system reliability
- Tested and integrated hardware components for critical pod safety monitoring, designing sensor interface topologies and implementing self-test procedures to verify sensor functionality and signal integrity on an STM32 using C/C++ and STM32CubeIDE
- Designed a custom **control PCB** using **Altium** to integrate embedded control hardware, power distribution, and sensor interfaces, consolidating multiple subsystems onto a single board to simplify system integration and improve maintainability

PROJECTS

Embedded Building Management System | *Python, Raspberry Pi, GPIO, LCD, DHT11, PIR*

June 2025

- Designed an embedded control system on a Raspberry Pi integrating DHT11 and PIR sensors to autonomously manage HVAC, lighting, fire alarm, and security subsystems via GPIO
- Implemented interrupt-driven event handling and an LCD-based interface with timestamped logging, enabling real-time monitoring and diagnostics of system status

TECHNICAL SKILLS

Languages: Python, C, C++, Java, Verilog, RISC-V, MIPS

Software/Tools: LabVIEW, Revit, AutoCAD, Altium, KiCad, Bluebeam, Arduino IDE, VSCode, Vivado, LTSpice, Cadence, Linux, Git, GitHub, STM32CubeIDE, MobaXterm

Hardware: Oscilloscope, Power Supply, Multimeter, Circuit Design & Analysis, Microcontrollers, ESP32, Embedded Systems, PCB Design, Arduino, Power Supplies, FPGA